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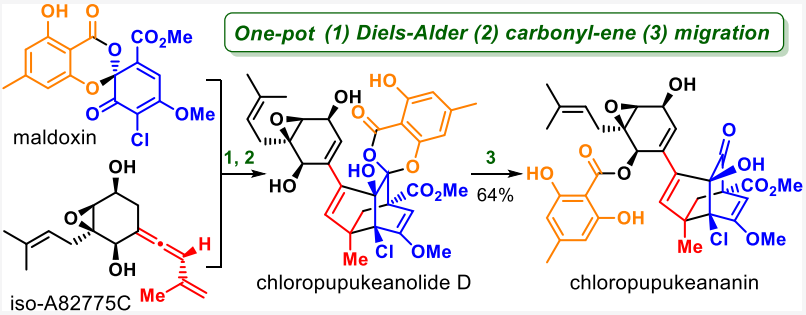
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Biomimetic Total Syntheses of (+)-Chloropupukeananin, (−)-Chloropupukeanolide D, and Chloropestolides

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ABSTRACT: Chloropupukeananin, chloropupukeanolides, and chloropestolides are a family of structurally complex bioactive natural products that possess highly functionalized tricyclo[4.3.1.0^{3,7}]decane or bicyclo[2.2.2]octane skeletons. Biosynthesis of the chloropupukeananin family is triggered by the intermolecular heterodimeric Diels−Alder reaction between maldoxin and iso-A82775C; however, the enzymes involved have not yet been identified. We herein report the one-pot biomimetic synthesis of chloropupukeananin and chloropupukeanolide D. Moreover, the effect of the solvent on the intermolecular Diels−Alder reaction of siccayne and maldoxin suggested that the biosynthesis of the chloropupukeananin family involves a Diels−Alderase-catalyzed heterodimeric Diels−Alder reaction.

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INTRODUCTION

The Diels−Alder reaction is one of the most powerful tools to introduce molecular complexity through biosynthetic[1] and chemical synthesis approaches.[2] Several natural products are synthesized through the Diels−Alder reaction in nature. However, most of these are only artifacts or products of spontaneous cycloaddition.[3] Since the discovery of a Diels−Alderase in the biosynthesis of solanapyrone,[4a] only a few classes of this enzyme, which can catalyze intramolecular reactions in polyketide synthesis,[4b−e] have been identified. Very recently, enzymes promoting the intermolecular Diels−Alder reactions to produce pseudo-dimeric resveratrols have been identified.[5] Artificial Diels−Alderases, such as antibody catalysts,[6] artificial enzymes,[7] and supramolecules,[8] have been developed to realize intermolecular Diels−Alder reactions owing to the high synthetic convergence and broad diversity of products of this process. If an intermolecular heterodimeric Diels−Alderase is identified, its genetic modifications should pave the way for the versatile production of bioactive natural product-like compounds.[9]

Chloropupukeananin (**1**), an inhibitor of HIV-1 replication, was isolated from the plant endophytic fungus *Pestalotiopsis fici* by Che *et al.* as the first chlorinated pupukeanane derivative (Figure 1).[10] Simultaneously, iso-A82775C (**2**) and pestheic acid (**3**) were also isolated and proposed as the biosynthetic precursors of **1**. One of the prominent structural features of **1** is its highly functionalized tricyclo[4.3.1.0^{3,7}]decane moiety that bears the vinyl allene moiety of **2** and the methyl benzoate

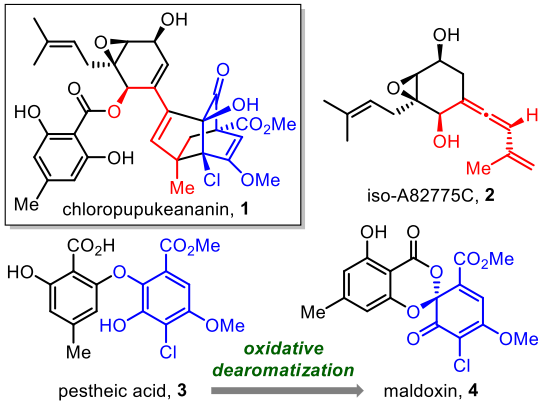


Figure 1. Chloropupukeananin and its biosynthetic precursors.

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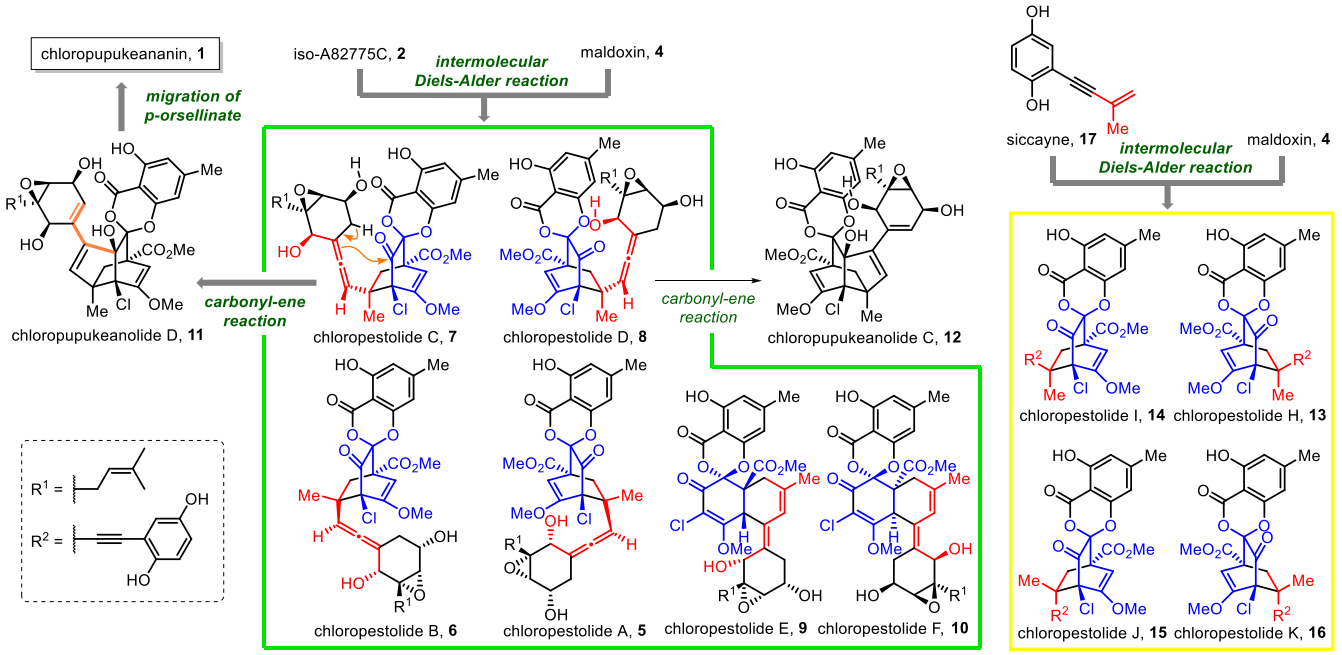


Figure 2. Proposed biosynthetic pathway of the chloropupukeananin family.

moiety of **3**. Thus, we proposed that (*R*)-maldoxin (**4**), which is generated by the asymmetric oxidative dearomatization of **3**, is the actual biosynthetic precursor instead of **3** (Figure 2).[11] Subsequently, an intermolecular reverse-electron-demand Diels–Alder reaction between **2** and **4** and an intramolecular carbonyl-ene reaction lead to the construction of the tricyclo[4.3.1.0^{3,7}]decane skeleton. Finally, migration of the *p*-orsellinate moiety results in **1**. Further efforts by Che *et al.* to unveil the biosynthetic pathway of **1** led to the identification of many natural heterodimeric congeners.[12] These included (i) chloropestolides A–F (**5**–**10**),[12a,b] the isomers of the Diels–Alder reactions, (ii) chloropupukeanolide D and C (**11**, **12**),[12c] a carbonyl-ene products of **7** and **8**, and (iii) other degradation products.[13] Finally, optically active (−)-(*R*)-**4**,[12d] which was the missing link in the biosynthesis of **1**, was isolated from the related fungus, *Pestalotiopsis theae*. These results support our hypothesis on the biosynthesis of **1**. This intriguing biosynthetic profile of **1** has attracted synthetic chemists, and the total syntheses of the biosynthetic precursors and the synthetic studies of **1** by us[14] and other groups[15] have been reported. Recently, chloropestolides H–K (**13**–**16**)[12e] were also isolated during the identification of the biosynthetic gene cluster of **2** using a prenyl transferase gene-disrupted strain. By inhibiting the biosynthesis of **2**, intermediate **17** was found to undergo intermolecular Diels–Alder reactions with **4** to afford **13**–**16**.

Isolation of the chloropestolides (**5**–**8** and **13**–**16**) indicates poor facial selectivity of the Diels–Alder reaction during biosynthesis (Figure 3). Briefly, during the formation of these bicyclo[2.2.2]octane-containing cycloadducts, the dienophiles approach the *Re*- or *Si*-face of C4′ of diene **4**, with *syn*- or *anti*-orientation between the R group of the dienophiles and the C2 acetal moiety of **4**. The intermolecular Diels–Alder reaction produces all the possible cycloadducts, with some *Si-syn* selectivity; these adducts appear to be artifacts owing to the poor selectivity. Thus, the chemical synthesis of a chloropupukeananin family through the Diels–Alder reaction between (+)-**2** and (−)-**4** can elucidate the biosynthetic

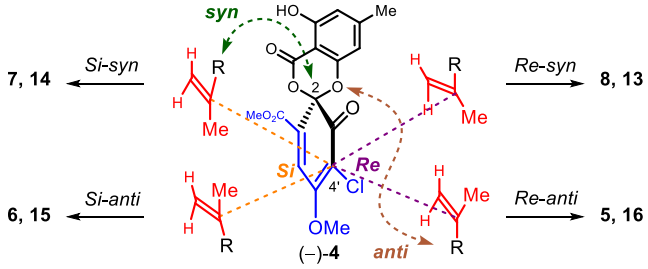


Figure 3. Four orientations of the diene and dienophile moieties in the intermolecular Diels–Alder reaction.

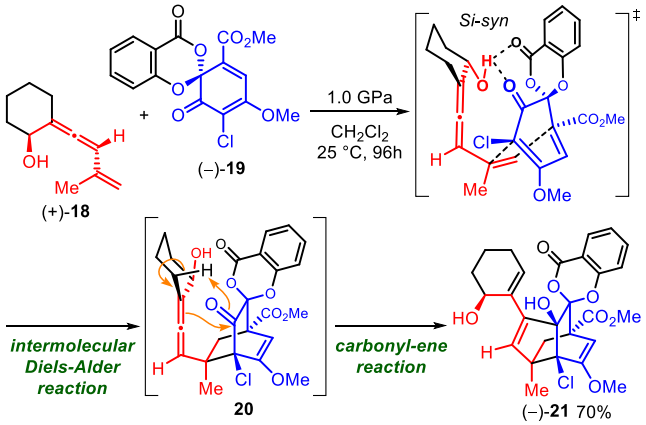
pathway, especially clarifying whether a nonenzymatic Diels–Alder reaction occurs or not. If an enzymatic reaction occurs, this enzyme might be the first example of an intermolecular heterodimeric Diels–Alderase, which could construct a highly functionalized bicyclo[2.2.2]octane skeleton. Thus, we herein report the biomimetic total synthesis of (+)-**1** from (+)-**2** and (−)-**4** using a one-pot Diels–Alder/carbonyl-ene cascade reaction and migration of the *p*-orsellinate group.[16]

RESULTS AND DISCUSSION

Our previous synthetic study on chloropupukeananin[14a] demonstrated the selective construction of a tricyclo-[4.3.1.0^{3,7}]decane skeleton under high-pressure conditions (Scheme 1). The Diels–Alder/carbonyl-ene cascade reaction of model compounds (+)-**18** and (−)-**19** results in tricyclic compound (−)-**21** in 70% yield *via Si-syn* cycloadduct **20**. The *Si-syn* transition state is stabilized by hydrogen-bonding networks. Therefore, the chemical Diels–Alder reactions between (+)-**2** and (−)-**4** should selectively produce *Si-syn* cycloadduct **7**, whereas that in nature exhibits poor selectivity, forming cycloadducts **5**–**10**. This is contradictory to the usual biosynthetic enzyme reactions. However, it is also possible that *Si-syn* **7** is biosynthetically obtained as a major product, without the involvement of enzymes. The subsequent reactions, such as the carbonyl-ene reaction or other

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Scheme 1. Diels–Alder/Carbonyl-ene Cascade Reaction in Our Previous Study



degradation reactions, consumed most of **7** in the presence or absence of enzymes. Consequently, all the cycloaddition products, **5**–**10**, are isolated in a nonselective manner. In addition, it is likely that the selectivity changes due to the effects of high-pressure conditions. Hence, a simple system to directly compare the cycloadducts of the chemical synthesis with the natural cycloadducts is required.

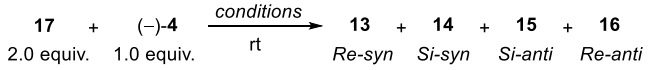
Accordingly, we investigated the biosynthetic Diels–Alder reaction of siccayne (**17**)[17] and (−)-**4** to produce **13**–**16** (Table 1). The reaction was initially conducted under several conditions that simulated the isolation procedure (entries 1–3). The reaction under neat conditions (entry 1) resulted in a mixture of products **13**–**16** (**13**/**14**/**15**/**16** yield ratio = 9:32:51:8) produced in a quantitative yield. Owing to the nature of (−)-**4**,[14c] the *Si*-face was preferred. Performing the reaction in EtOAc (extraction solvent, entry 2) and MeOH/H_2O (purification solvent, entry 3) resulted in a mixture containing predominant *Si-anti* **15**, which was incompatible with the *Si-syn* selectivity suggested by the ratio of the isolated natural products.[12e] Therefore, cycloadducts **13**–**16** were formed during cell cultivation. Surprisingly, the product ratio obtained in the aqueous solvent (entry 3) differed from that in the isolated sample; generally, hydrophilic solvents can simulate the biosynthetic reaction conditions.

Next, we considered the relationship between the product ratio and solvent to reproduce the isolated ratio of **13**–**16**. After investigating the Diels–Alder reaction in common organic solvents (entries 4–12; for the complete solvent screening data, see the Supporting Information), we established that the product ratio, especially for **14**/**15**, depended on the solvent basicity (SB)[18] and not on the other standard parameters, such as ε_r or μ. Since the SB scale represents the interaction between solvents and Brønsted acids, especially phenolic OHs, this result seems reasonable. In low-basicity solvents (SB < 0.2), **14** and **15** were obtained in a ratio of nearly 1:1. The best result for the *Si-syn* selectivity was obtained in CH_2Cl_2 (entry 6), giving **14** and **15** in isolated yields of 39 and 34%, respectively.[19] In contrast, in high-basicity solvents (SB > 0.2), the Diels–Alder reaction resulted in predominant **15**. The same trends were observed under high-pressure conditions (entries 13 and 14).

Comparing the ratio of **13**–**16** isolated from nature revealed that the preference for **14** was not reproduced. Comparatively similar results were obtained in lipophilic solvents (entries 5 and 6) rather than in hydrophilic solvents (entries 3, 11, and 12). This indicates that during the biosynthesis, the phenols have little interaction with surrounding molecules in the reaction environment, for example, a hydrophobic pocket. Thus, there was no possibility that cycloadducts **13**–**16** are artifacts during cell cultivation, extraction, and isolation. These results suggest that the biosynthetic production of the chloropupukeananin family involves the heterodimeric Diels–Alderase that catalyzed the intermolecular Diels–Alder reaction of (−)-**4**.

Subsequently, the biomimetic total synthesis of chloropupukeananin, based on our proposed biosynthetic pathway, was

Table 1. Intermolecular Diels–Alder Reaction between (−)-4 and 17

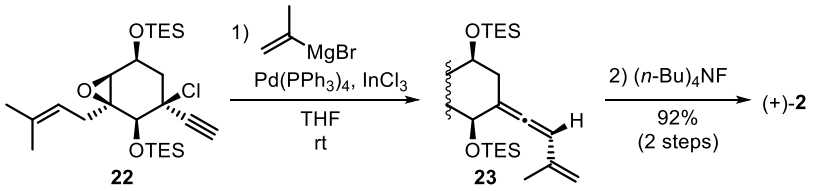


entry	conditions	SB	time (h)	ratio[a] 13	14	15	16
	isolation (ref 12e)			17	52	18	13
1	neat		24	9	32	51	8
2	EtOAc	0.542	48	6	16	71	7
3	MeOH/H_2O (4/1)		48	4	11	74	11
4	HFIP	0.014	48	14	37	34	15
5	benzene	0.124	48	9	42	44	5
6[b]	CH_2Cl_2	0.178	48	14	44 (39)	36 (34)	6
7	CH_3CN	0.286	48	10	28	53	9
8	acetone	0.475	48	6	20	64	10
9	MeOH	0.545	48	4	11	75	10
10	Et_2O	0.562	48	6	17	69	7
11	EtOH	0.658	48	3	7	80	10
12	*i*-PrOH	0.762	48	3	5	81	11
13[c]	MeOH/CH_2Cl_2 (10/1)		1	3	11	75	11
14[c]	CH_2Cl_2		1	15	46	33	6

[a]Ratio was determined by ^{1}H NMR. [b]Isolated yield in parentheses. [c]The reactions carried out using 1.1 equiv of **17** under high-pressure (1.0 GPa) conditions.

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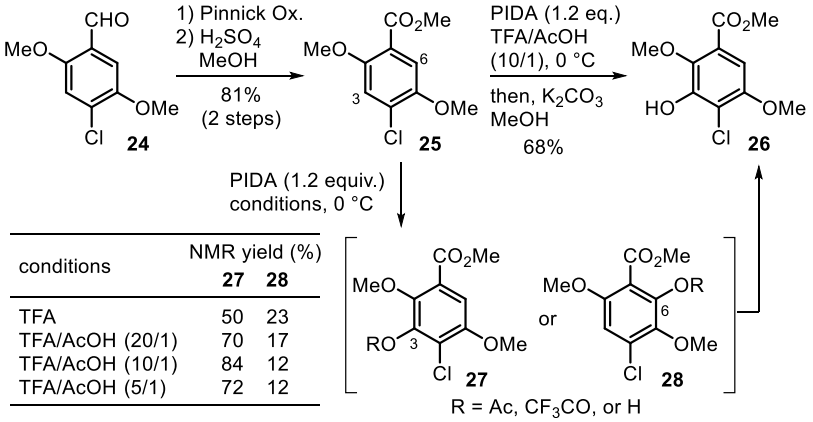
Table 2. Pd-Catalyzed *Anti*-S_N2' Reaction Using Organoindium Reagents



				NMR yield (%)		
entry	Grignard (equiv)	metal salt (equiv)	$Pd(PPh_3)_4$ (mol %)	**23**	**22**	chlorohydrins[a]
1[b]	5.0	CuCN (2.0)		30	59	
2	2.0	$ZnCl_2$ (2.0)		48	18	16
3	5.5	$InCl_3$ (2.0)	10	78		15
4	4.0	$InCl_3$ (2.0)	10	85		3
5	2.0	$InCl_3$ (2.0)	10	21	65	2
6	2.4	$InCl_3$ (1.2)	5.0	93		3

[a]Mixture of isomers; the structures were not identified. [b]Previous work (ref 14b). The reaction was carried out at −40 °C.

Scheme 2. Improved Synthesis of Phenol 26[a]



[a]PIDA = (diacetoxyiodo)benzene and TFA = trifluoroacetic acid.

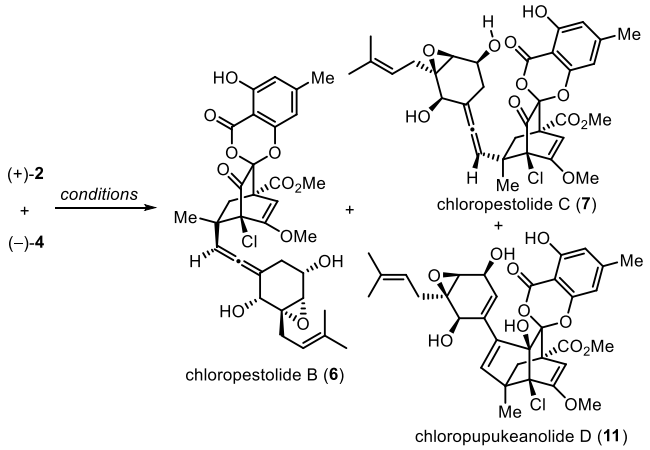
considered. Scaling up of the previously reported syntheses of (+)-**2**[14b] and (−)-**4**[14c] is associated with limitations, such as low conversion in the *anti*-S_N2' reaction of **23** using a vinyl copper reagent in the total synthesis of (+)-**2** (Table 2, entry 1). By using an organozinc reagent (entry 2) instead of the organocopper reagent, the conversion was slightly increased. However, a considerable amount of chlorohydrins were obtained as byproducts probably due to the Lewis acidity of $ZnCl_2$. This problem was addressed by using an organoindium reagent and Pd catalyst.[20] In the presence of $Pd(PPh_3)_4$, use of the organoindium reagent, fleshly prepared from isopropenyl−MgBr and $InCl_3$, resulted in a 78% yield of **23** (entry 3). The reactivity depended on the ratio of the reagents (entries 4 and 5); the best result was obtained by using 2:1 ratio of the Grignard reagent and $InCl_3$. Finally, the *anti*-S_N2' reaction using entry 6 and subsequently deprotecting the triethylsilyl groups of the resulting allene resulted in (+)-**2** in 92% yield on a subgram scale.

In the original synthesis of (−)-**4**, phenol **26** was prepared from 5-methoxysalicylic acid in five steps, one of which was a nonselective chlorination reaction.[11] Thus, the alternative synthesis of **26** was investigated (Scheme 2). Considering the directing effect of the aromatic electrophilic substitution, the preparation of phenol **26** should be carried out *via* a functionalization reaction at the C3 position of salicylate **25**. However, there are no examples of electrophilic substitutions that directly introduce hydroxyl groups into polysubstituted benzene rings. Therefore, we employed the Kita oxidative nucleophilic substitution reaction of phenyl ether using hypervalent iodine reagents.[21] Salicylate **25** was prepared by the Pinnick oxidation of **24**[22] and Fischer esterification in 81% yield. Under Muthukrishnan's conditions,[23] (diacetoxyiodo)-benzene (PIDA) in trifluoroacetic acid (TFA), the oxidative nucleophilic substitution reaction proceeded selectively at the C3 position (**27**/**28** = 2:1). Finally, we found that the use of a TFA/AcOH mixed solvent improved the selectivity to give **27** (**27**/**28** = 7:1), which underwent a one-pot hydrolysis to form **26** in 68% yield. Phenol **26** was converted to (−)-**4** according to our previously reported synthetic protocol.[12c]

Next, we investigated the Diels−Alder/carbonyl-ene cascade reaction using the biosynthetic precursors (+)-**2** and (−)-**4** (Table 3). A mixture of (+)-**2** and (−)-**4** was first subjected to high pressure in CH_2Cl_2, resulting in the formation of the desired *Si-syn* **7** and one diastereomer, *Si*-anti **6**, after 1 h (entry 1). Notably, no other cycloadducts were detected except for a small amount of **11**. The *Si-syn* selectivity would be attributed to the hydrogen-bonding networks, as shown in Scheme 1. Prolonging the reaction time facilitated the intramolecular carbonyl-ene reaction; when the reaction was conducted for 64 h (entry 4), **11** was obtained in 71% yield. Since the mixture of **7** and **11** was inseparable, it was heated under atmospheric pressure for the carbonyl-ene reaction to

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Table 3. Intermolecular Diels–Alder/Carbonyl-ene Cascade Reactions (+)-2 and (−)-4



					yield (%)[a]		
entry	ratio (2/4)	conditions	temp. (°C)	time (h)	6	7	11
1	1.3/1.0	1.0 GPa, CH_2Cl_2	25	1	17	67	3
2	1.3/1.0	1.0 GPa, CH_2Cl_2	25	4	17	62	13
3	1.3/1.0	1.0 GPa, CH_2Cl_2	25	16	17	42	40
4	1.3/1.0	1.0 GPa, CH_2Cl_2	25	64	17	5	71
5[b]	1.3/1.0	(i) 1.0 GPa, CH_2Cl_2, (ii) 0.1 MPa, neat	25, 60	64, 9	21		69
6	1.0/1.5	0.1 MPa, $(CH_2Cl)_2$	70	30	21	13	19
7[b]	1.0/1.0	(i) 0.1 MPa, neat,(ii) 0.1 MPa, neat	25, 60	120, 68	25		57

[a]NMR yield otherwise noted. [b]Isolated yield.

reach completion. After the high-pressure reaction (entry 5), the reaction mixture was concentrated and heated to 60 °C to afford **11** and **6** in isolated yields of 69 and 21%, respectively. We also investigated the Diels–Alder reaction under atmospheric pressure. The thermal reaction (entry 6) produced a mixture of **6**, **7**, and **11** in low yields owing to the decomposition of the starting materials. Next, the reaction was conducted under neat conditions (entry 7), followed by heating to 60 °C to produce the desired compound **11** in 57% yield, along with **6** in 25% yield. These results indicated that a high pressure significantly accelerated the Diels–Alder reactions between (+)-**2** and (−)-**4** but did not affect the facial selectivity. This synthetic protocol demonstrates the intriguing *Si*-face selectivity of the intermolecular Diels–Alder reaction between (+)-**2** and (−)-**4**; that is, only two cycloadducts, **6** and **7**, were obtained in the chemical synthesis, while six types of cycloadducts **5**–**10** were isolated as natural products.[12a,b] This also suggests that the biosynthetic intermolecular Diels–Alder reactions of **5**–**10** and **13**–**16** involve enzymes that activate the substrates to form diverse Diels–Alder products.

Finally, the migration of the *p*-orsellinate group of **11** was examined (Scheme 3a). We postulated that the secondary alkoxide group of **29** generated from **11** under basic conditions attacks the carbonyl carbon of the *p*-orsellinate group to form intermediate **30**. Then, the elimination reaction completes the migration to give **1**. Treatment of **11** with KO*t*-Bu in *N*,*N*-dimethylformamide (DMF) afforded **1** in 83% yield. Because the Diels–Alder/carbonyl-ene cascade reaction did not require any reagents, we attempted the one-pot biomimetic transformation from (+)-**2** and (−)-**4** (Scheme 3b). The atmospheric pressure cascade reaction (neat, 5 °C, 7 days and then 60 °C, 9 h) and the following migration reaction provided **1** in 64% yield and **6** in 20% yield, respectively. Thus, the first asymmetric total synthesis of **1** was accomplished. The spectral data of **1**, **6**, and **11**[24] were identical to those of the corresponding natural compounds.[10,12b,c]

CONCLUSIONS

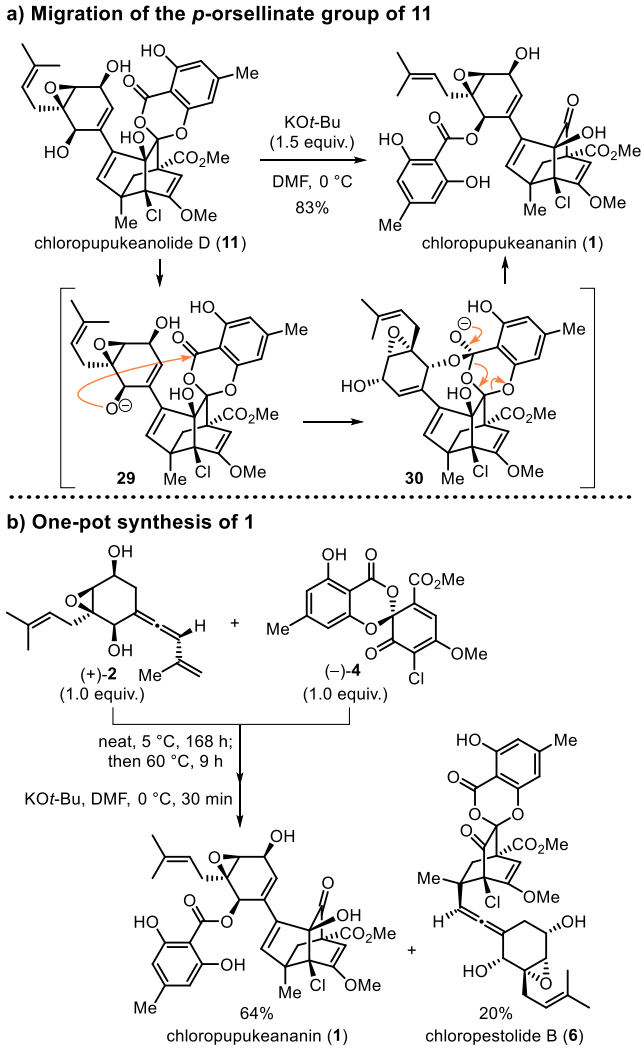
The biomimetic total synthesis of (+)-chloropupukeananin and (−)-chloropupukeanolide D through an intermolecular Diels–Alder reaction using (−)-maldoxin and (+)-iso-A82775C and a carbonyl-ene reaction was achieved. The biomimetic synthesis of chloropestolides H–K using (−)-maldoxin and siccayne unveiled the effect of the solvent on the intermolecular Diels–Alder reaction. The origin of the intermolecular Diels–Alder reaction stereoselectivity is still elusive; however, biosynthesis of the chloropupukeananin family possibly involves an enzymatically catalyzed intermolecular heterodimeric Diels–Alder reaction. We expected that our chemical synthesis of the family of chloropupukeananin helps the identification of the first intermolecular heterodimeric Diels–Alderase.

EXPERIMENTAL SECTION

General Information and Reagents. Infrared spectra were recorded using a JASCO FT-IR 4100 spectrometer with an ATR unit. Absorbance frequencies are recorded in reciprocal centimeters (cm^{-1}). High-resolution mass spectra (HRMS) were obtained from a Thermo Scientific Exactive for electrospray ionization (ESI). HRMS data are reported as *m*/*z* (relative intensity), with accurate mass reported for the molecular ion $[M + Na]^+$, $[M + H]^+$. ^1H and ^{13}C NMR spectra were recorded using a JEOL ECA-500 spectrometer operating at either 500 (^1H NMR) or 125 MHz (^{13}C NMR) in $CDCl_3$ as a solvent. Chemical shifts for NMR were reported in parts per million relative to the chemical shift of the residual solvent (^1H NMR 7.26 ppm for $CDCl_3$ and 2.05 ppm for acetone-d_6 and ^{13}C NMR 77.0 ppm for $CDCl_3$, 39.5 ppm for DMSO-d_6, and 29.8 ppm for acetone-d_6). Multiplicities are indicated as br (broad), s (singlet), d

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Scheme 3. Completion of Biomimetic Total Synthesis of Chloropupukeananin[a]



[a]DMF = *N*,*N*-dimethylformamide.

(doublet), t (triplet), q (quartet), or m (multiplet). Coupling constants (*J*) are reported in Hertz (Hz). All reactions sensitive to oxygen or moisture were performed under an atmosphere of dry argon in a flame-dried glassware. Chemical reagents and anhydrous solvents were of commercial grade and were used without any purification. 2-Chloro-1,4-dimethoxybenzene, hexamethylenetetramine, and 2-methyl-2-butene were purchased from Tokyo Chemical Industry Co., Ltd. Siccayne (**17**),[17] (+)-**2**,[14b] and (−)-**4**[14c] were prepared by the known procedures. All reactions that require heating were carried out using an oil bath. Hyperbaric reactions were carried out in a high-pressure apparatus (PVS-1400R, TERAMECS Co., Ltd., Japan). Flash chromatography was performed with silica gel 60N (spherical, neutral), purchased from KANTO Chemical Industries Ltd, unless otherwise noted. Analytical thin-layer chromatography (TLC) was performed using commercial silica gel plates (E. Merck, silica gel 60 F_{254}).

General Procedure for Table 1, Entries 2−12. Siccayne (34.8 mg, 0.200 mmol) and (−)-maldoxin (**4**) (38.0 mg, 0.100 mmol) were stirred in each solvent (0.50 mL) for 48 h at room temperature. Then, the reaction mixture was concentrated *in vacuo* to afford a mixture of chloropestolides H−K (**13**−**16**). The ratio of **13**−**16** was calculated by ^{1}H NMR. When using CH_2Cl_2 as a solvent (entry 6), the residue was purified by silica gel column chromatography (hexane/EtOAc = 2/1 to 1/1) to afford a mixture of **13**−**16** (54.8 mg, 99%, **13**/**14**/**15**/**16** = 13:44:37:6). The mixture was further purified by preparative TLC (benzene/acetonitrile = 9/1) to afford **14** (21.9 mg, 39%), **15** (18.9 mg, 34%), and an inseparable mixture of **13** and **16** (9.3 mg, 17%).

*(−)-Chloropestolide I (**14**).* White solid; R_f 0.48 (benzene/acetonitrile = 4/1); mp 122−124 °C; ^{1}H NMR (acetone-d_6, 500 MHz): δ 9.83 (1H, br), 7.89 (1H, s), 7.27 (1H, s), 6.82 (1H, t, J = 1.7 Hz), 6.73−6.71 (2H, m), 6.52 (1H, s), 6.32 (1H, s), 5.76 (1H, s), 3.84 (3H, s), 3.76 (3H, s), 3.28 (1H, d, J = 13.2 Hz), 2.30 (3H, s), 2.19 (1H, d, J = 13.2 Hz), and 1.49 (3H, s); ^{13}C{^{1}H} NMR (acetone-d_6), 125 MHz: δ 190.1, 168.9, 163.1, 161.5, 155.9, 152.0, 151.7, 150.9, 150.2, 119.4, 118.5, 117.0, 112.6, 110.5, 108.2, 98.0, 97.8, 97.0, 95.1, 81.9, 81.0, 57.2, 53.4, 51.9, 42.8, 42.0, 26.9, and 22.3; IR (ATR) ν_{max}: 3503, 3102, 2957, 2929, 2850, 1762, 1706, 1638, 1581, 1496, 1458, 1361, 1317, 1239, 1200, 1149, 1100, and 1041 cm^{-1}; HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{28}H_{23}O_{10}ClNa$, 577.0872; found, 577.0869; $[\alpha]_D^{27}$ −137.1 (*c* 0.18, MeOH).

*(−)-Chloropestolide J (**15**).* White solid; R_f 0.54 (benzene/acetonitrile = 4/1); mp 124−127 °C; ^{1}H NMR (acetone-d_6, 500 MHz): δ 9.78 (1H, br), 7.92 (1H, br), 7.33 (1H, br), 6.75−6.72 (3H, m), 6.53 (1H, s), 6.33 (1H, s), 5.79 (1H, s), 3.84 (3H, s), 3.75 (3H, s), 2.87 (1H, d, J = 13.7 Hz), 2.68 (1H, d, J = 13.7 Hz), 2.30 (3H, s), and 1.54 (3H, s); ^{13}C{^{1}H} NMR (acetone-d_6, 125 MHz): δ 190.9, 168.8, 162.9, 161.5, 155.7, 152.0, 151.9, 151.8, 150.9, 118.9, 118.4, 117.0, 112.7, 110.4, 108.3, 98.2, 97.8, 96.8, 96.7, 80.9, 79.4, 57.3, 52.0, 43.8, 43.0, 25.2, and 22.3; IR (ATR) ν_{max}: 3503, 3105, 2979, 2953, 2937, 2842, 1761, 1708, 1639, 1580, 1495, 1460, 1315, 1283, 1234, 1199, 1148, and 1113 cm^{-1}; HRMS (ESI) m/z: [M + Na]$^+$ calcd for $C_{28}H_{23}O_{10}ClNa$, 577.0872; found, 577.0870; $[\alpha]_D^{27}$ −79.2 (*c* 0.2, MeOH).

*Mixture of Chloropestolides H and K (**13** and **16**).* White solid; R_f 0.51 (benzene/acetonitrile = 4/1); ^{1}H NMR (acetone-d_6, 500 MHz): δ 9.79 (1H, br), 9.73 (1H, br), 7.92 (1H, br), 7.90 (1H, br), 7.71 (1H, br), 7.36 (1H, br), 6.79 (1H, d, J = 8.6 Hz), 6.76−6.72 (4H, m), 6.60 (1H, d, J = 2.9 Hz), 6.57 (1H, s), 6.51 (1H, s), 6.45 (1H, s), 6.20 (1H, s), 5.75 (1H, s), 5.75 (1H, s), 3.83 (3H, s), 3.81 (6H, s), 3.79 (3H, s), 3.45 (1H, d, J = 13.2 Hz), 3.03 (1H, d, J = 13.2 Hz), 2.71 (1H, d, J = 13.8 Hz), 2.34 (3H, s), 2.17 (1H, d, J = 13.2 Hz), 2.01 (3H, s), 1.60 (3H, s), and 1.49 (3H, s); ^{13}C{^{1}H} NMR (acetone-d_6, 125 MHz): δ 189.4, 188.6, 168.79, 168.76, 164.3, 163.6, 161.5, 161.1, 154.3, 154.1, 152.1, 151.93, 151.92, 151.6, 151.4, 150.94, 150.86, 150.2, 119.4, 118.9, 118.4, 117.2, 117.0, 113.2, 112.9, 110.7, 110.5, 109.3, 108.3, 98.9, 98.4, 97.8, 97.7, 97.3, 96.9, 95.6, 81.4, 81.3, 79.5, 57.4, 57.3, 53.5, 51.7, 51.5, 44.0, 43.8, 43.1, 40.9, 27.1, 25.4, 22.3, and 22.0.

(+)-Iso-A82775C (Table 2, Entry 6). Preparation of a solution of organoindium reagent [in tetrahydrofuran (THF), 0.15 M]: to a suspension of dried $InCl_3$ (177 mg, 0.80 mmol) in THF (2.2 mL) was added a solution of isopropenyl magnesium bromide in THF (0.5 M, 3.2 mL, 1.6 mmol) at 0 °C. The mixture was warmed up to room temperature and stirred for 30 min.

To a solution of **22** (282 mg, 0.58 mmol) and $Pd(PPh_3)_4$ (35 mg, 0.030 mmol, 5.0 mol %) in THF (4.0 mL) was added the freshly prepared solution of organoindium reagent in THF (0.15 M, 4.7 mL, 0.71 mmol, 1.2 equiv) at 0 °C, and the mixture was stirred for 5 h at room temperature. The reaction mixture was quenched with saturated aqueous NH_4Cl solution. The aqueous layer was extracted with *n*-hexane. The combined organic layers were dried over $MgSO_4$, filtrated, and concentrated *in vacuo* to yield the crude allene **23** using for the next step without further purification. To a solution of the crude allene **23** in THF (7.0 mL) was added a solution of tetrabutylammonium fluoride in THF (1.0 M, 1.60 mL, 1.60 mmol, 3.0 equiv) at 0 °C, and the mixture was stirred for 30 min. The reaction mixture was quenched with brine, and the aqueous layer was extracted with EtOAc. The combined organic layers were dried over $MgSO_4$, filtrated, and concentrated *in vacuo*. The residue was purified by flash silica-gel column chromatography (*n*-hexane/EtOAc = 3/1) to provide iso-A82775C (**2**) (140 mg, 92%) as a colorless oil. The analytical data are identical to those reported in our previous paper.[14b]

*4-Chloro-2,5-dimethoxybenzaldehyde (**24**).* To a mixture of 2-chloro-1,4-dimethoxybenzene (4.2 mL, 30 mmol) and hexamethyle-

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netetramine (4.2 g, 30 mmol) was added TFA (45 mL) at room temperature, and the mixture was heated to reflux for 14 h. The hot solution was poured onto crushed ice and added solid $NaHCO_3$ (70 g) cautiously. The yellow solid was formed and filtered through a Buchner funnel. The solid was washed with water and hexane and then dissolved with EtOAc. The organic layer was washed with water, dried over $MgSO_4$, filtrated, and concentrated *in vacuo* to afford **24** (5.2 g, 87%) as a yellow solid. The analytical data are identical to those reported in the literature.[22]

24: yellow solid; R_f 0.38 (*n*-hexane/EtOAc = 4:1 v/v); mp 99−100 °C (*n*-hexane/EtOAc); ^{1}H NMR ($CDCl_3$, 500 MHz): δ 10.39 (1H, s), 7.38 (1H, s), 7.07 (1H, s), 3.90 (3H, s), and 3.90 (3H, s).

4-Chloro-2,5-dimethoxybenzoic Acid. To a solution of **24** (2.01 g, 10.0 mmol) in THF (20 mL), *t*-BuOH (20 mL), H_2O (10 mL), and 2-methyl-2-butene (10 mL, 100 mmol, 10 equiv) were added $NaH_2PO_4{\cdot}2H_2O$ (9.4 g, 60 mmol, 6.0 equiv) and $NaClO_2$ (2.7 g, 30 mmol, 3.0 equiv) at 0 °C. The mixture was stirred for 1 h at the same temperature. To the reaction mixture was added H_2O and extracted with EtOAc. The combined organic layers were concentrated *in vacuo*. The white solid was dissolved with NaOH aqueous solution. The aqueous layer was washed with Et_2O twice. The aqueous layer was acidified to pH 1 and extracted with EtOAc. The combined organic layers were dried over $MgSO_4$, filtrated, and concentrated *in vacuo* to afford crude 4-chloro-2,5-dimethoxybenzoic acid as a white solid using the next step without further purification.

4-Chloro-2,5-dimethoxybenzoic Acid. R_f 0.12 (EtOAc);^{1}H NMR ($CDCl_3$, 500 MHz): δ 10.72 (1H, br s), 7.73 (1H, s), 7.13 (1H, s), 4.06 (3H, s), and 3.93 (3H, s); $^{13}C\{^1H\}$ NMR (DMSO-d_6, 125 MHz): δ 166.6, 152.5, 148.1, 125.4, 120.3, 115.1, 114.5, 56.7, and 56.5; IR (ATR) ν_{max}: 2988, 2949, 2870, 1685, 1498, 1379, 1270, 1211, 1142, and 1027 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + Na]^+$ calcd for $C_9H_9O_4ClNa$, 239.0082; found, 239.0085.

*Methyl 4-Chloro-2,5-dimethoxybenzoate (**25**).* To a solution of crude 4-chloro-2,5-dimethoxybenzoic acid in MeOH (30 mL) was added concentrated H_2SO_4 (1.5 mL) dropwise, and the solution was heated to reflux for 1.5 h. The reaction mixture was cooled to room temperature, diluted with H_2O, concentrated *in vacuo*, and extracted with EtOAc. The organic layer was washed with aqueous $NaHCO_3$, dried over $MgSO_4$, filtered, and concentrated *in vacuo* to afford **25** (1.86 g, 81%, 2 steps) as a white solid.

25; R_f 0.23 (*n*-hexane/EtOAc = 4:1 v/v); mp 76−77 °C (*n*-hexane/EtOAc);^{1}H NMR ($CDCl_3$, 500 MHz): δ 7.42 (1H, s), 7.04 (1H, s), 3.902 (3H, s), 3.897 (3H, s), and 3.87 (3H, s); $^{13}C\{^1H\}$ NMR ($CDCl_3$, 125 MHz): δ 165.6, 153.4, 148.4, 127.4, 118.2, 114.8, 114.7, 56.6, 56.5, and 52.0; IR (ATR) ν_{max}: 2927, 2946, 2844, 1719, 1573, 1498, 1466, 1440, 1375, 1327, 1204, 1183, 1097, 1022, and 766 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + Na]^+$ calcd for $C_{10}H_{11}O_4ClNa$, 253.0238; found, 253.0243.

*Methyl 4-Chloro-3-hydroxy-2,5-dimethoxybenzoate (**26**).* To a solution of **25** (1.04 g, 4.5 mmol) in TFA (10 mL) and AcOH (1.0 mL) was added PIDA (1.74 g, 5.4 mmol, 1.2 equiv) at 0 °C, and the mixture was stirred for 1 h at the same temperature. The reaction mixture was basified to pH 9 by K_2CO_3 aqueous solution and added MeOH (10 mL). The mixture was stirred for 1 h at 70 °C. The reaction mixture was neutralized, concentrated *in vacuo*, and extracted with EtOAc. The organic layer was washed with aqueous $NaHCO_3$, dried over $MgSO_4$, filtered, and concentrated *in vacuo*. The residue was purified by flash silica-gel column chromatography (*n*-hexane/EtOAc = 10/1 to 3/1) to provide **26** (752 mg, 68%) as a white solid. The analytical data are identical to those reported in our previous paper.[11]

26; R_f 0.16 (*n*-hexane/EtOAc = 4:1 v/v); mp 116−118 °C (*n*-hexane/EtOAc); ^{1}H NMR (500 MHz, $CDCl_3$): δ 6.97 (s, 1H), 6.14 (br, 1H), 3.94 (s, 3H), 3.91 (s, 3H), and 3.90 (s, 3H); $^{13}C\{^1H\}$ NMR (125 MHz, $CDCl_3$): δ 165.4, 151.6, 147.4, 142.0, 121.5, 113.7, 103.9, 62.5, 56.4, and 52.5.

General Procedure for High-Pressure Conditions (Table 3, Entries 1−4). A mixture of (−)-**4** (11.5 mg, 0.030 mmol) and (+)-iso-A82775C (**2**) (10.4 mg, 0.040 mmol) was placed in a 1.0 mL polytetrafluoroethylene (PTFE) vial which was then filled with CH_2Cl_2 (1.0 mL). The vial was tightly closed and kept in a high-pressure apparatus at 1.0 GPa at room temperature for several hours. After depressurizing, the solvent was removed *in vacuo*. The crude mixture was purified by flash silica-gel column chromatography (hexane/AcOEt = 3/1 to 2/1) to afford a mixture of chloropestolide B (**6**), chloropestolide C (**7**), and chloropupukeanolide D (**11**). The yield was calculated from ^{1}H NMR.

Procedure for High-Pressure/Atmospheric Pressure Conditions (Table 3, Entry 5). A mixture of (−)-**4** (76.4 mg, 0.20 mmol) and (+)-**2** (66.3 mg, 0.25 mmol) was placed in a 1.0 mL PTFE vial which was then filled with CH_2Cl_2 (1.0 mL). The vial was tightly closed and kept in a high-pressure apparatus at 1.0 GPa at room temperature for 64 h. After depressurizing, the solvent was removed *in vacuo*. The mixture was stirred for 9 h at 60 °C. After the full consumption of **7** was checked by ^{1}H NMR, the mixture was directly purified by flash silica-gel column chromatography (*n*-hexane/EtOAc = 2/1) to provide **11** (88.0 mg, 69%) as a white solid and **6** (26.8 mg, 21%) as a white solid.

Procedure for Atmospheric Pressure Conditions (Table 3, Entry 7). A mixture of (−)-**4** (37.9 mg, 0.10 mmol) and (+)-**2** (26.2 mg, 0.10 mmol) was placed in a sealed tube. Then, to make a homogeneous mixture, the mixture was dissolved in a small amount of CH_2Cl_2, and CH_2Cl_2 was removed in vacuo using a rotary evaporator. The mixture was placed for 120 h at room temperature. After the full consumption of **2** and **4** was checked by ^{1}H NMR, the mixture was stirred for 68 h at 60 °C. After the full consumption of **7** was checked, the mixture was directly purified by flash silica-gel column chromatography (*n*-hexane/EtOAc = 2/1) to provide **11** (36.5 mg, 57%) and **6** (16.0 mg, 25%).

*(−)-Chloropupukeanolide D (**11**).* R_f 0.30 (*n*-hexane/EtOAc = 1:1 v/v); mp 132−133 °C (*n*-hexane/EtOAc); ^{1}H NMR (acetone-d_6, 500 MHz): δ 9.95 (1H, s), 6.65 (1H, s), 6.39 (1H, s), 6.38 (1H, s), 6.09 (1H, br s), 5.48 (1H, s), 5.43 (1H, s), 5.20 (1H, m Hz), 4.35 (1H, d, *J* = 10.4 Hz), 4.19 (1H, br s), 3.70−3.68 (4H, m), 3.67 (3H, s), 3.44 (1H, d, *J* = 10.9 Hz) 3.29 (1H, s), 2.87 (1H, dd, *J* = 14.9, 8.7 Hz), 2.55 (1H, d, *J* = 13.2 Hz) 2.27 (3H, s), 2.24 (1H, dd, *J* = 14.9, 6.3 Hz), 1.74 (3H, s), 1.68 (3H, s), 1.47 (1H, d, *J* = 13.2 Hz), and 1.14 (3H, s); $^{13}C\{^1H\}$ NMR (acetone-d_6, 125 MHz): δ 170.4, 164.1, 160.9, 157.7, 153.2, 151.0, 140.9, 140.7, 135.3, 131.9, 129.0, 119.2, 110.8, 108.4, 108.1, 97.0, 96.0, 94.2, 89.4, 68.2, 65.8, 64.8, 60.4, 56.3, 55.4, 52.8, 47.4, 42.3, 32.3, 25.9, 22.6, 22.4, and 18.0; IR (ATR) ν_{max}: 3422, 2955, 2925, 2856, 1736, 1691, 1639, 1582, 1459, 1367, 1260, 1204, 1164, and 1017 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + Na]^+$ calcd for $C_{33}H_{35}O_{11}ClNa$, 665.1760; found, 665.1767; $[\alpha]_D^{24}$ −3.8 (*c* 0.12, MeOH).

*(+)-Chloropestolide B (**6**).* R_f 0.21 (*n*-hexane/EtOAc = 1:1 v/v); mp 102−104 °C (*n*-hexane/EtOAc); ^{1}H NMR (acetone-d_6, 500 MHz): δ 9.81 (1H, s), 6.53 (1H, s), 6.31 (1H, s), 5.63 (1H, s), 5.20−5.16 (2H, m), 4.23−4.18 (2H, m), 4.01−3.96 (2H, m), 3.79 (3H, s), 3.76 (3H, s), 3.26 (1H, s), 2.82 (1H, dd, *J* = 14.6, 9.2 Hz), 2.56 (1H, d, *J* = 13.7 Hz), 2.38−2.33 (1H, m), 2.31−2.28 (4H, m), 2.15 (1H, dd, *J* = 12.0 5.2 Hz), 2.09 (1H, dd, *J* = 14.6, 5.2 Hz), 1.76 (3H, s), 1.66 (3H, s), and 1.22 (3H, s); $^{13}C\{^1H\}$ NMR (acetone-d_6, 125 MHz): δ 203.1, 191.4, 169.3, 163.1, 161.4, 155.7, 151.8, 151.1, 135.6, 119.3, 112.6, 108.2, 105.8, 98.3, 97.2, 96.9, 96.8, 81.7, 68.6, 68.4, 65.5, 63.1, 57.0, 53.4, 51.5, 44.9, 39.9, 33.4, 31.1, 25.9, 23.9, 22.3, and 18.1; IR (ATR) ν_{max}: 2923, 2852, 1739, 1706, 1626, 1580, 1464, 1438, 1359, 1277, 1201, and 1022 cm^{-1}; HRMS (ESI) *m*/*z*: $[M + Na]^+$ calcd for $C_{33}H_{35}O_{11}ClNa$, 665.1760; found, 665.1774; $[\alpha]_D^{24}$ +5.2 (*c* 0.10, MeOH).

(+)-Chloropupoukeananin (1). To a solution of **11** (24.0 mg, 0.037 mmol) in DMF (1.0 mL) was added a solution of *t*-BuOK in THF (1.0 M, 0.055 mL, 0.055 mmol, 1.5 equiv) at 0 °C, and the mixture was stirred for 30 min at the same temperature. The reaction mixture was quenched with 1 M HCl aqueous solution, and the aqueous layer was extracted with EtOAc. The combined organic layers were dried over $MgSO_4$, filtrated, and concentrated *in vacuo*. The residue was purified by flash silica-gel column chromatography (*n*-hexane/EtOAc = 2/1) to provide **1** (20.0 mg, 83%).

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*One-Pot Procedure from (+)-**2** and (−)-**4**.* A mixture of (−)-**4** (47.0 mg, 0.124 mmol) and (+)-**2** (32.5 mg, 0.124 mmol) was placed in a sealed tube. Then, to make a homogeneous mixture, the mixture was dissolved in a small amount of CH_2Cl_2, and CH_2Cl_2 was removed in vacuo using a rotary evaporator. The mixture was placed for 168 h in a refrigerator (5 °C). After the full consumption of **2** and **4** was checked by ^{1}H NMR, the mixture was heated at 60 °C for 9 h. To the mixture were added DMF (3.0 mL) and a solution of *t*-BuOK in THF (1.0 M, 0.18 mL, 0.18 mmol, 1.5 equiv) at 0 °C and stirred for 30 min at the same temperature. The reaction mixture was quenched with 1 M aqueous HCl solution, and the aqueous layer was extracted with EtOAc. The combined organic layers were dried over $MgSO_4$, filtrated, and concentrated *in vacuo*. The residue was purified by flash silica-gel column chromatography (*n*-hexane/EtOAc = 2/1) to provide **1** (51.0 mg, 64%) and **6** (15.9 mg, 20%).

*(+)-Chloropupoukeananin (**1**).* R_f 0.30 (*n*-hexane/EtOAc = 1:1 v/v); mp 117−118 °C (*n*-hexane/EtOAc);^{1}H NMR (acetone-d_6, 500 MHz): δ 9.74 (2H, br), 6.44 (1H, s), 6.37 (1H, s), 6.29 (2H, m), 6.00 (1H, s), 5.61 (1H, d, J = 2.9 Hz), 5.51 (1H, s), 5.16 (1H, t, J = 7.5 Hz), 4.56 (1H, m), 4.44 (1H, d, J = 7.5 Hz), 3.76 (3H, s), 3.67 (3H, s), 3.49 (1H, t, J = 2.9 Hz) 2.68 (1H, d, J = 12.1 Hz), 2.52 (1H, dd, J = 15.0, 7.5 Hz), 2.40 (1H, dd, J = 15.0, 7.5 Hz) 2.25 (3H, s), 1.59 (3H, s), 1.56 (1H, d, J = 12.1 Hz), 1.54 (3H, s), and 1.03 (3H, s); ^{13}C{^{1}H} NMR (acetone-d_6, 125 MHz): δ 196.3, 170.8, 169.1, 161.9, 156.1, 149.2, 139.0, 138.7, 136.1, 133.1, 125.6, 118.2, 109.4, 98.5, 92.8, 88.4, 86.5, 70.8, 65.3, 62.4, 61.3, 61.0, 56.6, 52.8, 48.8, 48.7, 33.0, 25.7, 21.9, and 18.0; IR (ATR) ν_{max}: 3440, 2925, 2857, 1745, 1725, 1660, 1642, 1571, 1449, 1330, 1300, 1261, 1232, 1188, and 1092 cm^{-1}; HRMS (ESI) m/z: $[M + Na]^+$ calcd for $C_{33}H_{35}O_{11}ClNa$, 665.1760; found, 665.1766; $[\alpha]_D^{21}$ +33.2 (*c* 0.12, MeOH).

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.1c02108.

Summary of our total syntheses of (−)-**4** and (+)-**2**; entire solvent screening data for the Diels−Alder reaction of (−)-**4** and **17**; NMR comparison of our synthetic **1**, **6**, **11**, and **13**−**16**; and ^{1}H and ^{13}C NMR spectra of all new compounds (PDF)

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Notes

The authors declare no competing financial interest.

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